# PROJECT DESIGN PHASE

## SOLUTION ARCHITECTURE

**Date:** 25 September 2026
**Team ID:** SWTID-2026-4307
**Project Name:** AI WeatherWise API
**Maximum Marks:** 4 Marks

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## Solution Architecture

Solution architecture is a process that connects the project problem with the proposed technology solution. It describes the structure, components, data flow, features, and requirements of the **AI WeatherWise API** system.

The goals of the solution architecture are to:

- Provide real-time weather information to users.
- Obtain weather data through a weather API.
- Process the received weather information.
- Provide useful weather insights and recommendations.
- Display important weather details in a simple format.
- Define the major components and their responsibilities.
- Describe how data flows between the user, application, API, and output.

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## Architecture Components

### 1. User

The user enters a city or location to obtain weather information.

### 2. AI WeatherWise Application

The application receives the user's request and sends the required location information to the weather API.

### 3. Weather API

The weather API provides weather information for the requested location.

### 4. AI / Processing Module

The received weather data is processed to organize the information and generate useful weather insights.

### 5. Weather Output

The processed information is displayed to the user, including relevant weather details and insights.

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## Data Flow

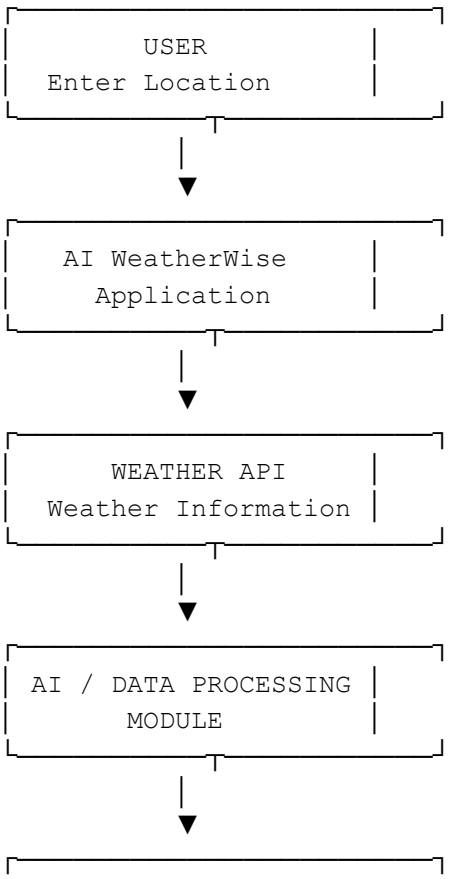
The basic data flow of the AI WeatherWise API is:

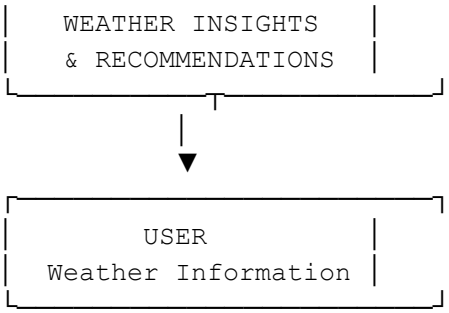
**User → AI WeatherWise Application → Weather API → Weather Data → AI / Processing Module → Weather Insights → User**

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## Solution Architecture Diagram

### Figure 1: Architecture and Data Flow of AI WeatherWise API






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## System Requirements

### Functional Requirements

- User should be able to enter a location.
- System should retrieve weather information.
- System should process the received weather data.
- System should display weather information.
- System should provide useful weather insights.

### Technical Requirements

- Weather API
- Application/API layer
- Data processing module
- User interface
- Internet connection

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## Expected Solution

The **AI WeatherWise API** provides a structured solution for obtaining weather information, processing the data, and presenting useful weather insights to users. The solution architecture defines the major components and explains the flow of information between the user, application, weather API, processing module, and final output.

**Figure 1: Architecture and Data Flow of AI WeatherWise API**

**Reference:** AI WeatherWise API Project Documentation